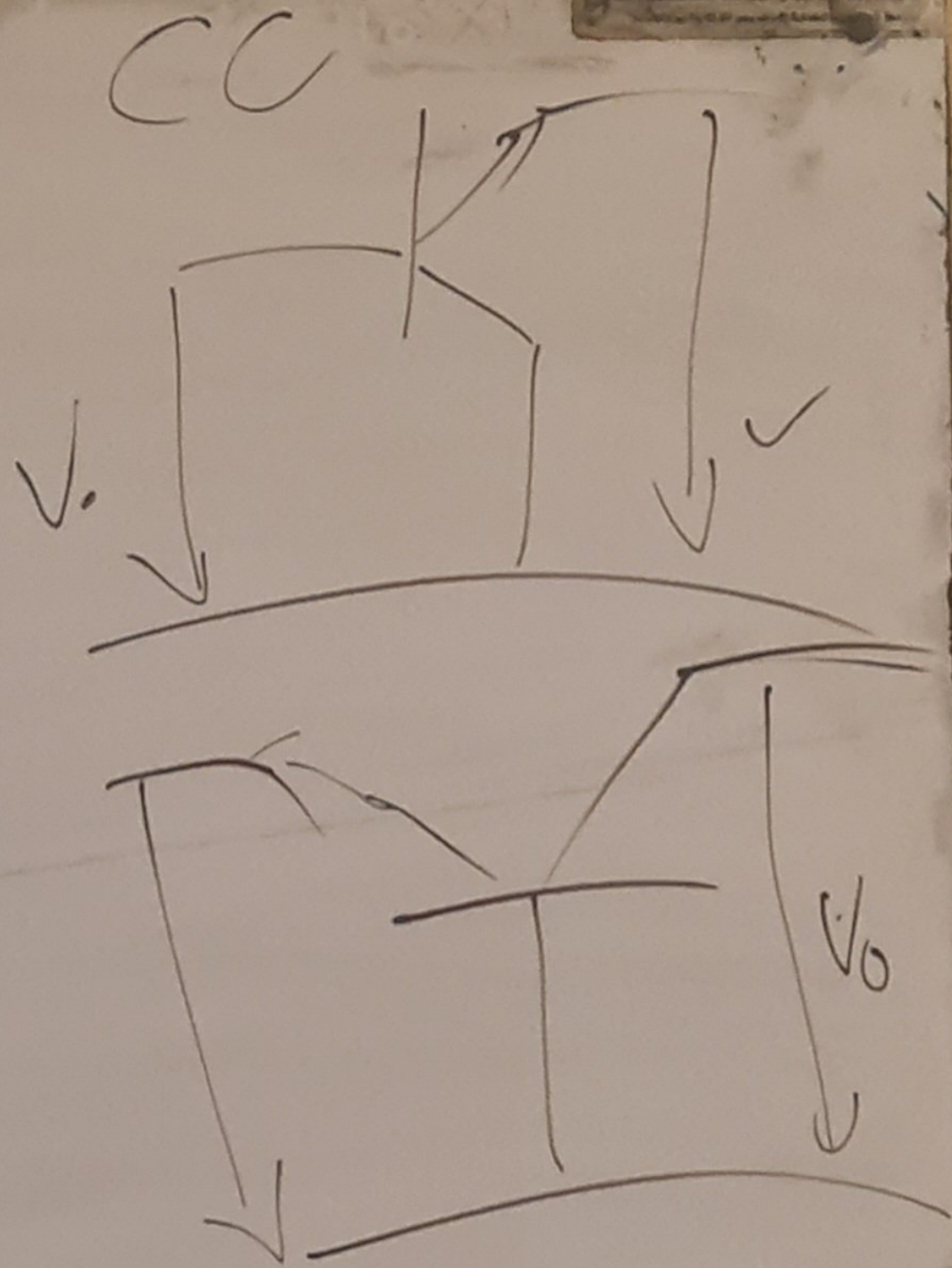
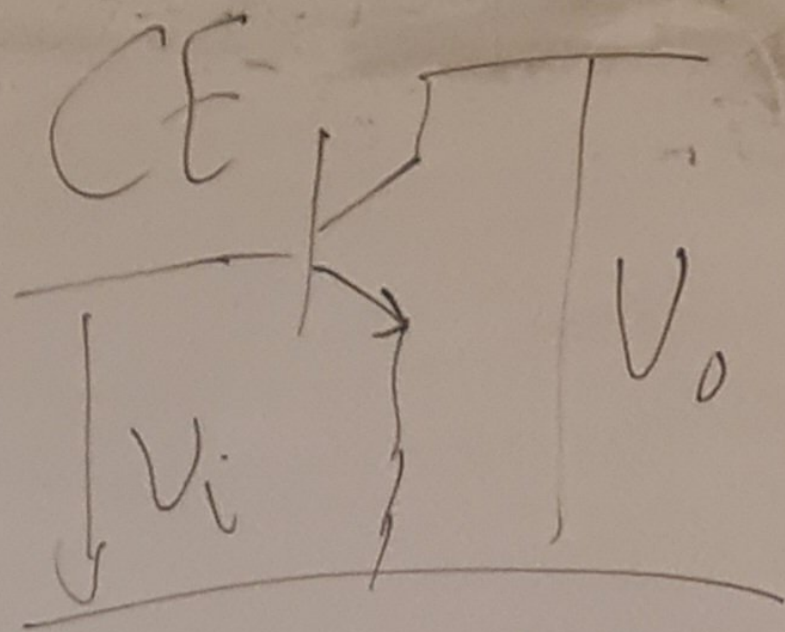
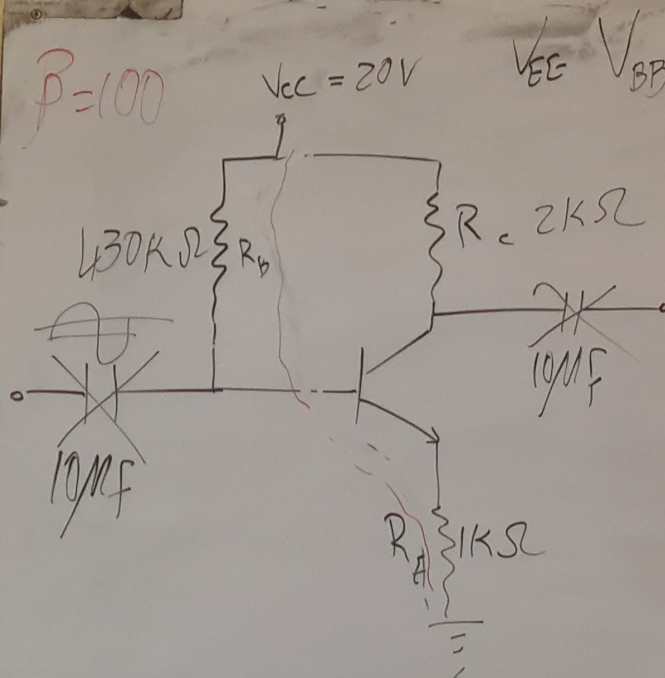




$$\beta = 100$$



sol $\sum V = 0$

$$20 - V_B - V_{BE} - V_E = 0$$

$$20 - I_B R_B - 0.7 - I_E R_E = 0 \Rightarrow I_B R_B + I_E R_E = 19.3$$

$$I_B R_B + I_B (\beta + 1) R_E = 19.3$$

$$I_B \{ R_B + (\beta + 1) R_E \} = 19.3$$

$$\beta = \frac{I_C}{I_B}$$

$$I_E = I_C + I_B$$

$$I_C = \beta I_B$$

$$I_E = \beta I_B + I_B$$

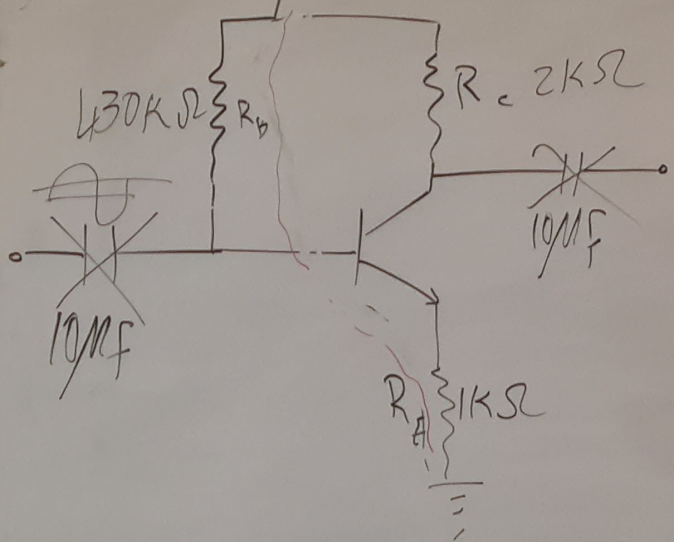
$$\Rightarrow I_E = I_B (\beta + 1)$$

$$I_B = \frac{19.3}{R_B + (\beta + 1) R_E} = \frac{19.3}{430 \times 10^3 + 101 + 1000}$$

$$I_B = 0.000044 \text{ A}$$

$$\beta = 100$$

$V_{CC} = 20V$ V_{EE} V_{BB} V_{SS} V_{DD}



sol $\sum V = 0$

$$20 - \frac{V}{B} - \frac{V}{BE} - \frac{V}{E} = 0$$

$$20 - I_B R_B - 0.7 - I_E R_E = 0 \Rightarrow I_B R_B + I_E R_E = 19.3$$

$$I_B R_B + I_B (\beta + 1) R_E = 19.3$$

$$I_B \{ R_B + (\beta + 1) R_E \} = 19.3$$

$$\beta = \frac{I_C}{I_B}$$

$$I_E = I_C + I_B$$

$$I_C = \beta I_B$$

$$I_E = \beta I_B + I_B$$

$$\Rightarrow I_E = I_B (\beta + 1)$$

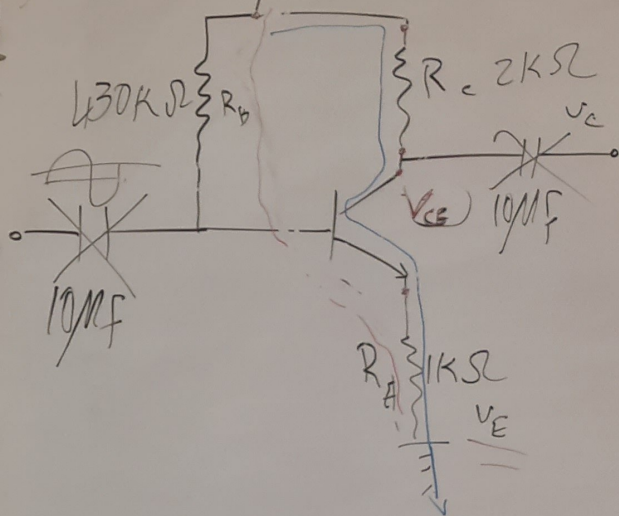
$$I_B = \frac{19.3}{R_B + (\beta + 1) R_E} = \frac{19.3}{430 \times 10^3 + 101 + 1000}$$

$$I_B = 0.000044 A$$

$$\beta = 100$$

$$V_{CC} = 20V$$

$$V_{EE} \quad V_{BB} \quad V_{SS} \quad V_{DD}$$



$$V_{CE} = 20 - I_B \{ \beta R_C + (\beta + 1) R_E \}$$

$$\Rightarrow V_{CE} = 20 - 44 \times 10^{-6} \{ 100 \times 2000 + 101 \times 1000 \}$$

sol $\sum V = 0$

$$20 - V_C - V_{CE} - V_E = 0$$

$$20 - I_C R_C - V_{CE} - I_E R_E = 0$$

$$I_C R_C + V_{CE} + I_E R_E = 20$$

$$\left(\beta \frac{I_B}{\beta} \right) R_C + V_{CE} + \frac{I_B}{\beta} (\beta + 1) R_E = 20$$

$$\frac{I_B}{\beta} \{ \beta R_C + (\beta + 1) R_E \} = 20$$

$$\beta = \frac{I_C}{I_B}$$

$$I_E = I_C + I_B$$

$$I_C = \beta I_B$$

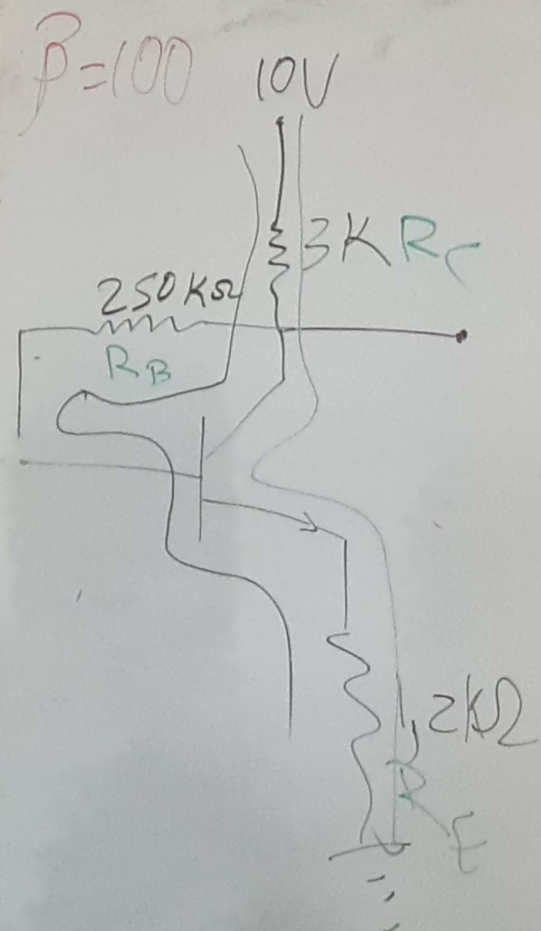
$$I_E = \beta I_B + I_B$$

$$I_E = I_B (\beta + 1)$$

$$I_B = \frac{19.3}{R_B + (\beta + 1) R_E} = \frac{19.3}{430 \times 10^3 + 101 + 1000}$$

$$I_B = 0.000044 A$$

$$44 \mu A$$



$$\sum V = 0$$

$$10 - V_C - V_B - V_{BE} - V_E = 0$$

$$10 = I_C R_C + I_B R_B + V_{BE} + I_E R_E$$

$$10 = I_B R_C + I_B R_B + 0.7 + I_B (\beta + 1) R_E$$

$$9.3 = I_B (R_C + R_B + (\beta + 1) R_E)$$

$$I_B = \frac{9.3}{R_C + R_B + (\beta + 1) R_E}$$

$$= \frac{9.3}{100(3 \times 10^3) + 250 \times 10^3 + 101 + 1.2 \times 10^3}$$

$$\beta = \frac{I_C}{I_B}$$

$$I_E = I_C + I_B$$

$$I_E = I_C + I_B$$

$$I_B = \frac{9.3}{R_B + (\beta + 1) R_E} = \frac{9.3}{430 \times 10^3}$$

$$\beta = 100$$

10V

$$10 - V_C - V_{CE} - V_C = 0$$

$$10 = R_C I_C - V_{CE} - I_C R_C$$

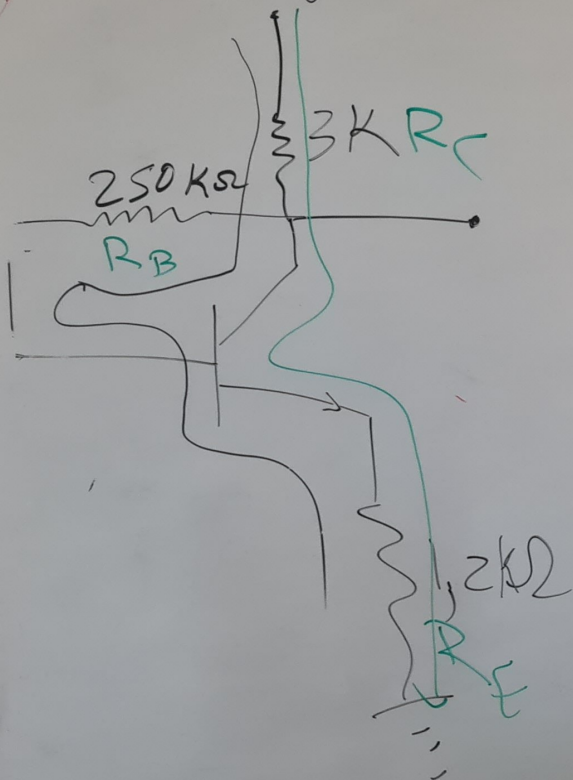
$$\beta = \frac{I_C}{I_B}$$

$$I_E = I_C + I_B$$

$$= \beta I_B + I_B$$

$$I_E = \beta I_B + I_B$$

$$I_E = I_B (\beta + 1)$$



$$I_B = \frac{9.3}{\beta R_C + R_B + (\beta + 1) R_E}$$

$$= \frac{9.3}{100(3 \times 10^3) + 250 \times 10^3 + 101 + 1.2 \times 10^3}$$

$$I_B = \frac{9.3}{R_B + (\beta + 1) R_E} = \frac{9.3}{430 \times 10^3 + 101 + 1000}$$

$$I_B = 0.000044 \text{ A}$$

$$I_B = 44 \mu\text{A}$$